

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GLY 422 — Past Questions & Materials
by Renance

Course: GLY 422
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — GLY 422 PQ 2024

Level 400 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
DEPARTMENT OF EARTH SCIENCES
SECOND SEMESTER EXAMINATION 2023/2024 SESSION
COURSE: GLY 422 (ENGINEERING GEOLOGY 2UNITS) TIME: 2HRS

Answer question 1 and any other 2 questions

(1) A soil sample is collected from the field and weighed with a container, the weight of the wet soil or sample and the container is 94.98g, and after oven drying, the weight of oven dried soil and container is 83.43g, if the volume of the container used is 256.76cm³

(ai) Calculate the bulk density of the sample (5marks)

(aii) Assuming the particle size or densities (PS) of the soil is 1.99g/cm³, calculate the porosity of the soil sample (5marks)

(1b) Assuming that the moisture content of a sample collected from the field weighs 25g is 28%. Calculate the weight of the dry sample, if the mass of water is 5g.(7 marks)

(1c) From the table below, fill in the missing figures and plot the water content as against the nos. Of blows and determine the liquid limit.(8 marks)

SAMPLE NUMBER	1	2	3	4
MOISTURE CONTENT	11	1	5	4
Mc= mass of empty can	22.23	23.31	21.87	22.28
McMs= mass of can + moist soil	28.56	29.27	25.73	25.72
McMds = mass of can + dry soil	27.40	28.10	24.90	24.60
Ms = mass of solid	5.19	4.79	3.03	2.32
Mw = mass of pore water	1.16	1.17	0.83	0.62
W = water content %	26.18	24.43	27.39	30.69
Number of blows	31	29	20	14

(1d) What is the significance of liquid and plastic limits in soil classification and explain how these values are used to calculate the consistency index.(5marks)

(2a) The issue of surface and groundwater poses a lot of threats to some engineering structures. Critically discuss in details the influence of these factors on some engineering structures. (10marks).

Feasibility, site selection, design, specification and construction are key words in terms development of dams. Discuss. (10marks).

(3a) Discuss in detail the influence of foundation to the safety of dams. (10marks)

(3b). Discuss the two most important mechanical/design properties of the foundation materials that must be computed. (5marks)

(3c) The design of any foundation is generally consist of three basic operations. Discuss (5marks)

(4a). Geologist play a critical role in addressing engineering problems in terms of roads, dams, and bridges. Highlight the applications to these problems. (15marks)

(4b). List the principles of a site investigation work (5marks)

(5a). Discuss some the environmental impacts of a dam. (5marks)

(5b). Define Atterberg limit and explain the purpose of determining the consistency in soil engineering (5marks)

(5c). Highlight the importance of Atterberg limit in road construction. (10marks)

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